MIP_REASONING_ENGINE.md

Mainframe Intelligence Platform

Enterprise Reasoning Engine Architecture

Version: 1.0

Status: Core Intelligence Layer

Classification: Strategic Platform Component

Purpose

The Reasoning Engine transforms:

Metadata

+

Knowledge Graph

↓

Insights

↓

Recommendations

↓

Modernization Intelligence

Without reasoning:

```
Repository → Metadata → Graph
```

With reasoning:

```
Repository
    ↓
```

```
Metadata
     ↓
Knowledge Graph
     ↓
Reasoning Engine
     ↓
Insights
     ↓
Recommendations
     ↓
Transformation Roadmap
```

Executive Vision

The Reasoning Engine answers:

Understanding

What does this program do?

Impact

What breaks if this changes?

Risk

How dangerous is this component?

Ownership

Which capability owns this?

Modernization

What should we modernize first?

Transformation

What should become a service?

Design Principles

Evidence Driven

Every conclusion must reference evidence.

Explainable

Every recommendation must be traceable.

Confidence Based

Every inference receives confidence.

Deterministic First

Rule-based reasoning before LLM reasoning.

Human Verifiable

Architects must be able to validate outputs.

Reasoning Layers

```
Layer 1
Metadata Reasoning

Layer 2
Graph Reasoning

Layer 3
Business Reasoning

Layer 4
Modernization Reasoning

Layer 5
Strategic Reasoning
```

Layer 1

Metadata Reasoning

Purpose:

Generate insights directly from metadata.

Example:

Program:

TS2AUTH

Metrics:

LOC = 12,000

Complexity = 85

Dependencies = 97

Inference:

HIGH COMPLEXITY

Inputs

```
Program Metadata

Job Metadata

DB2 Metadata

Dataset Metadata
```

Outputs

```
Complexity Score

Risk Indicators

Ownership Indicators
```

Layer 2

Graph Reasoning

Purpose

Infer relationships from graph structure.

Example

```
TS2AUTH

↓

CARD_MASTER

↓

SETTLEMENT

↓

BILLING
```

Inference

```
Critical Transaction Path
```

Algorithms

```
Centrality

Shortest Path

Connected Components

Community Detection

Graph Traversal
```

Outputs

```
Critical Nodes

Critical Edges

Execution Paths

Dependency Chains
```

Layer 3

Business Reasoning

Purpose

Translate technical assets into business meaning.

Example

Programs

```
AUTHDRV
AUTHVAL
AUTHPOST
```

Inference

```
Authorization Capability
```

Inputs

```
Program Clusters

Transactions

Tables

Business Rules
```

Outputs

```
Capabilities

Domains

Business Processes
```

Layer 4

Modernization Reasoning

Purpose

Generate modernization opportunities.

Example

```
Capability

Authorization
```

Graph shows:

```
High Cohesion

Low External Coupling
```

Inference

```
Microservice Candidate
```

Inputs

```
Capabilities

Dependencies

Data Ownership

Execution Flows
```

Outputs

```
Service Candidates

API Candidates

Event Candidates
```

Layer 5

Strategic Reasoning

Purpose

Create transformation plans.

Example

```
Capability Criticality = HIGH

Complexity = HIGH

Risk = HIGH
```

Inference

```
Modernize Later
```

Outputs

```
Migration Waves

Roadmaps

Risk Mitigation Plans

Investment Priorities
```

Core Reasoning Engines

Impact Analysis Engine

Questions

```
What breaks if TS2AUTH changes?
```

Output

```
Affected Programs

Affected Jobs

Affected Tables

Affected Datasets

Affected Capabilities
```

Root Cause Engine

Questions

```
Where did this failure originate?
```

Output

```
Execution Chain

Root Component

Dependency Path
```

Service Discovery Engine

Questions

```
What should become a service?
```

Output

```
Service Candidates

Boundaries

Dependencies
```

Capability Discovery Engine

Questions

```
What business capability does this support?
```

Output

```
Capability Name

Programs

Transactions

Tables
```

Risk Assessment Engine

Questions

```
How risky is modernization?
```

Output

```
Risk Score

Risk Factors

Mitigation Strategy
```

Confidence Model

Every conclusion receives confidence.

Scale

```
0.0 - 1.0
```

Example

```
Program Name

1.0

Capability Assignment

0.82

Service Candidate

0.71
```

Explainability Model

Every recommendation includes:

```
{
  "recommendation": "",

  "confidence": 0.91,

  "evidence": [],

  "graph_paths": [],

  "reasoning_steps": []
}
```

Recommendation Model

Types

```
SERVICE_CANDIDATE

API_CANDIDATE

EVENT_CANDIDATE

DEAD_CODE

HIGH_RISK

WAVE_1

WAVE_2

WAVE_3

WAVE_4
```

Modernization Readiness Score

Formula

```
Metadata Quality

+

Graph Quality

+

Lineage Completeness

+

Impact Confidence

+

Capability Coverage
```

Produces

```
0–100
```

Example Reasoning Flow

Question

```
What should be modernized first?
```

Reasoning

```
Capability Analysis

↓

Dependency Analysis

↓

Risk Analysis

↓

Complexity Analysis

↓

Modernization Scoring

↓

Wave Assignment
```

Output

```
Wave 1 Candidates

Wave 2 Candidates

Wave 3 Candidates

Wave 4 Candidates
```

Future AI Integration

Phase 1

```
Rule Based Reasoning
```

Phase 2

```
Graph Assisted Reasoning
```

Phase 3

```
LLM Assisted Reasoning
```

Phase 4

```
Multi-Agent Reasoning
```

Google ADK Integration

Future Agents

```
Discovery Agent

Parser Agent

Metadata Agent

Graph Agent

Lineage Agent

Impact Agent

Modernization Agent

Architecture Agent
```

Each agent contributes evidence to the reasoning engine.

Enterprise Scale Targets

Support

```
100,000+ Programs

20,000+ Jobs

10M+ Relationships

Thousands of Capabilities
```

Success Criteria

An architect can ask:

```
What is TS2AUTH?

Why is it important?

What depends on it?

What business capability owns it?

What modernization path is recommended?
```

And receive an explainable, evidence-backed answer without reading COBOL.

MIP Principle

Metadata stores facts.

Knowledge Graph stores relationships.

Reasoning Engine creates understanding.

Understanding enables transformation.

End of Document.